Data Engineering in Large Language Model Pretraining: A Survey

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***Abstract*—As large language models (LLMs) grow, the demand for high-quality pretraining text makes efficient use of available data increasingly important. Drawing on public technical reports and open corpus projects, this survey examines data engineering for text-based LLM pretraining. Following the progression from corpus construction to data utilization and evaluation, we review practices supported by large-scale training and distinguish them from methods whose transfer to frontier models remains uncertain. A common mathematical formulation connects document processing, sampling distributions, token-level objectives, and evaluation under resource constraints. We synthesize recurring practices and disagreements concerning deduplication, filtering, synthetic augmentation, mixture optimization, and staged training. Throughout, we interpret findings in light of incomplete public disclosure and differences in experimental protocols. The survey provides a reference for interpreting reported practices and identifying unresolved questions in pretraining data engineering.**

***Index Terms*—Data engineering, large language models, pretraining**

I. Introduction

Large language models (LLMs) acquire language representations and generative capabilities through pretraining on large text corpora. Empirical scaling laws relate improvements in predictive performance to model size, training data, and computation [1, 2]. As training scales up, the number of tokens consumed has also grown. GPT-3 [3] used approximately 300B tokens, whereas DeepSeek V3 [4] and LLaMA 3 [5] report 14.8T and 15.6T tokens, respectively. Qwen 3 [6], GLM-5 [7], and DeepSeek V4 [8] report budgets in the tens of trillions. Comparing these budgets requires consistent tokenization and accounting for repeated exposure.

Continued expansion faces constraints in both data supply and data utilization. Web corpora contain duplicated pages, extraction artifacts, and heterogeneous content, so raw volume does not directly measure usable training information. Villalobos et al. [9] project that demand may approach the available stock of human-generated public text between 2026 and 2032 under their stated assumptions. This projection does not establish a date at which usable text will be exhausted. Longpre et al. [10] document increasing access restrictions among high-quality web sources, while provenance, consent, and copyright complicate the use of otherwise accessible material [11]. These constraints motivate more efficient learning from admissible text.

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How efficiently models learn from available text provides a second motivation. Human learning offers a useful point of reference, although differences in sensory input, interaction, and objectives prevent direct efficiency comparisons with LLMs [12]. Data engineering addresses this question through acquisition, cleaning, selection, mixing, and scheduling. RefinedWeb [13] demonstrates the effectiveness of processed web data relative to curated mixtures in its evaluation setting. FineWeb-Edu [14] and DCLM [15] further show that filtering choices can materially affect performance under controlled training budgets. Together, these results make data processing a design choice alongside architecture and compute.

Existing surveys approach this subject from complementary perspectives. Albalak et al. [16] review data selection, while Wang et al. [17] examine data management across pretraining and supervised fine-tuning. Related work surveys the interaction between LLMs and data management [18], and efficiency methods spanning models, data, and training systems [19]. This survey retains a narrower focus on text-based pretraining and on practices supported by large-scale training. Public reports for GPT, LLaMA, Qwen, DeepSeek, Kimi, GLM, OLMo, and Phi, together with open corpus projects, form the principal evidence base. We organize these observations into a common pipeline, compare choices within each stage, and identify promising methods whose adoption or effectiveness at frontier scale remains unresolved.

We distinguish reported practices from controlled findings because adoption alone does not isolate a contribution, and omission may reflect limited disclosure. The disclosure inventory in Appendix A retains contextual evidence from broader model reports, but excludes post-training and multimodal disclosures from text-pretraining evidence.

The methodological discussion also incorporates recent studies available through October 13, 2026, including preprints on acquisition, filtering, synthesis, scheduling, and evaluation. These studies extend the model inventory with methods whose frontier-scale relevance still requires experimental validation.

The survey follows the progression from the pretraining objective to corpus construction, utilization, and evaluation. Section II formalizes pretraining from a data-engineering perspective. Section III reviews corpus construction, and Section IV examines how a corpus is converted into training signals. Section V discusses evaluation and comparability. Section VI synthesizes the shared pipeline and empirical consensus, and Section VII concludes with future directions.

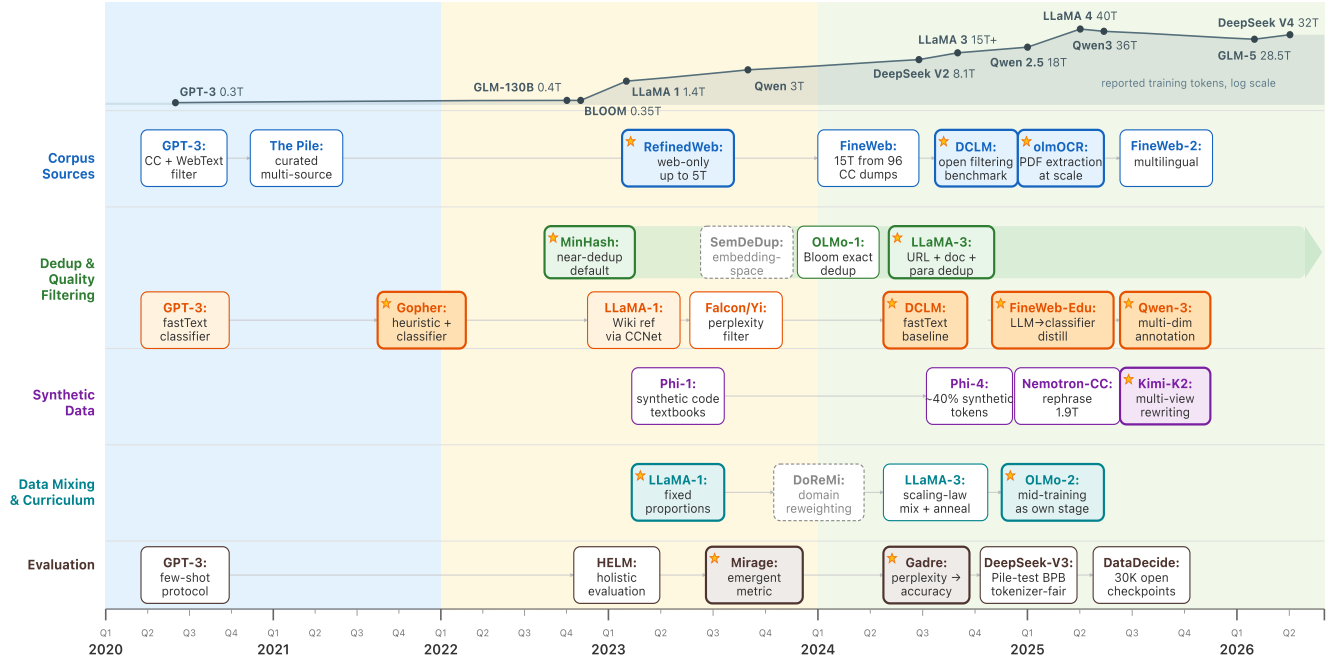


Fig. 1. Selected developments in pretraining data engineering. The timeline summarizes the practices discussed in this survey without implying uniform adoption or comparable training budgets.

II. Formulating LLM Pretraining

Data engineering determines which text is available for pretraining and how it contributes to learning. We formalize this process through a constructed corpus, a data utilization policy, and a prediction objective, providing common notation for the subsequent sections.

A. Corpus and Training Exposure

Definition 1 (Corpus and training exposure)**.** *Let $\mathcal{R} = \{r_i\}_{i=1}^{n}$ be an indexed collection of raw documents. A construction procedure $\mathcal{C}$ produces a tokenized corpus $\mathcal{D} = \mathcal{C}(\mathcal{R}) = \{x_i\}_{i=1}^{m}$, where $x_i = \tau(r_i')$, τ is a fixed tokenizer with vocabulary $\mathcal{V}$, and r_i' is a retained or transformed document. The stored token count is $U = \sum_{i=1}^{m} |x_i|$. Training exposure B counts non-padding token presentations consumed during training, including repeated presentations of the same stored tokens.*

Thus, U measures stored volume and B training consumption, neither counting distinct semantic facts. Partial traversal or repeated sampling means that B need not equal U. We use $\mathcal{D}_k$ for source or domain stratum k, with K strata in total. Comparing overlapping strata requires explicit assignment or weighting conventions.

B. Data Utilization and the Pretraining Objective

Definition 2 (Data utilization policy)**.** *At training step s, the policy $\pi_s = (q_s, \mathcal{T}_s, L_s, w_s)$ specifies a document-sampling distribution q_s, a sequence transformation $\mathcal{T}_s$, a maximum context length L_s, and a loss-weighting rule w_s. After sampling and transformation, position t in sequence i has context $c_{i,t}$ and target token $x_{i,t}$. The rule w_s assigns a nonnegative coefficient $w_{s,i,t}$ to the loss for predicting that target.*

Sampling controls which documents are presented, while w_s controls how their target losses are combined. These coefficients are neither sampling probabilities nor model parameters. Cropping, packing, and rearrangement determine the prediction contexts within the limit L_s. Fig. 2 illustrates these choices for a fixed corpus.

We represent the LLM by conditional probabilities $p_\theta(v \mid c)$ over $v \in \mathcal{V}$, with parameters θ. For ordinary next-token pretraining, $c_{i,t} = x_{i,<t}$ and

$$p_\theta(x_i) = \prod_{t=1}^{|x_i|} p_\theta(x_{i,t} \mid x_{i,<t}). \tag{1}$$

The empty prefix is represented by an initial context. This formulation captures the text-prediction objective used in GPT-3, PaLM, and LLaMA [3, 20, 21]. The same notation accommodates other contexts produced by data transformations.

For a batch $\mathcal{B}_s$ of transformed sequences, let $\ell_{i,t}(\theta) = -\log p_\theta(x_{i,t} \mid c_{i,t})$. The empirical objective is

$$Z_s = \sum_{i \in \mathcal{B}_s} \sum_{t} w_{s,i,t},$$
$$\widehat{\mathcal{L}}_s(\theta) = \frac{1}{Z_s} \sum_{i \in \mathcal{B}_s} \sum_{t} w_{s,i,t} \ell_{i,t}(\theta), \tag{2}$$

where $Z_s > 0$ normalizes the weights within the batch. Unit weights on valid targets recover token-averaged cross-entropy. A zero weight omits a target loss without necessarily removing the token from later contexts. Positive weights set relative contributions: weights of 2 and 1 give the first target twice the loss coefficient of the second. Uniformly rescaling positive weights does not change this objective.

The final model results from the sequence of objectives induced by $\mathcal{C}$ and $\{\pi_s\}$. When the policy changes during

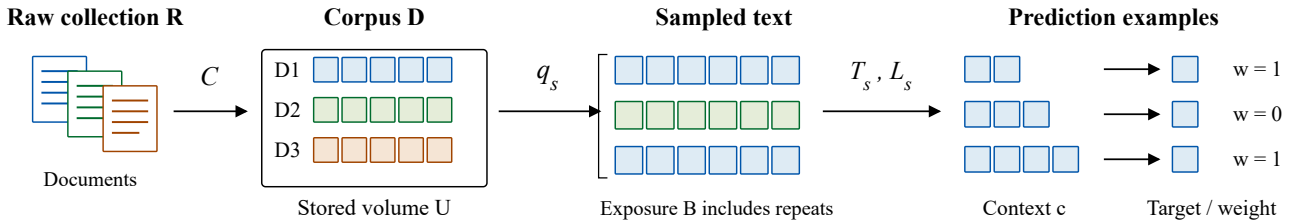


Fig. 2. A data-centric view of pretraining. Construction determines the available corpus, sampling controls exposure, and presentation defines prediction contexts and targets. Colored tokens indicate source domains. Repeated selections illustrate why stored volume U and training exposure B differ.

Table I. Notation used throughout the survey.

Symbol	Meaning
$\mathcal{R}, \mathcal{D}, \mathcal{D}_k$	Raw collection, constructed corpus, domain stratum
U, B, F	Stored tokens, training exposure, training compute
P, θ	Parameter count and model parameters
$q_s, \alpha_{s,k}$	Document distribution and domain sampling weight
$\mathcal{T}_s, L_s$	Sequence transformation and context bound
$c_{i,t}, w_{s,i,t}$	Prediction context and target-loss coefficient
π_s, η_s	Data utilization policy and learning rate
$\mathcal{E}, \Gamma$	Evaluation data and evaluation protocol

training, the model need not optimize a single fixed corpus objective. Sections III and IV examine how corpus and utilization choices affect these objectives.

C. Resource Constraints and Comparison

Data decisions must be interpreted relative to resource constraints. If b_s is the number of non-padding token presentations at step s, then $B = \sum_s b_s$. For dense models, a frequently used training-compute approximation is $F \approx 6PB$, where P denotes the parameter count [1, 2]. This approximation omits context-dependent costs and implementation overheads, and should not be applied unchanged to every model. Processing, synthesis, and mixture search incur additional costs.

Comparisons of data recipes should specify the controlled resources and assess the resulting models under common evaluation data $\mathcal{E}$ and protocol Γ (Section V). Table I collects the notation connecting construction, utilization, and evaluation.

III. Corpus Construction

Corpus construction turns raw documents into training-ready text through the transformation $\mathcal{D} = \mathcal{C}(\mathcal{R})$ in Definition 1. We review data curation (§III-A), deduplication (§III-B), quality filtering (§III-C), and synthetic augmentation (§III-D), while distinguishing the text retained in the corpus from its subsequent use in training.

A. Data Curation

Earlier public corpora often combined multiple curated sources. The Pile [22] aggregates 22 English sources totaling approximately 825 GiB, and CLUECorpus2020 [23] provides approximately 100 GB of Chinese text. C4 [24] establishes a reproducible approach to filtering CommonCrawl. Curated sources broaden coverage but have limited scale. Repeated CommonCrawl snapshots supply larger volumes with substantial page and temporal overlap, requiring extraction, normalization, and provenance tracking.

Processed web data can be competitive with curated mixtures under controlled comparisons. RefinedWeb [13] demonstrates this result using a web-focused pipeline. FineWeb [14], DCLM [15], Dolma [25], and Nemotron-CC [26] subsequently provide different approaches to extraction, filtering, and corpus release. Table II compares evaluation settings and distinguishes corpus scale from training exposure.

Recent corpus releases extend this approach along two complementary dimensions. FineWeb2 adapts segmentation, filtering, and deduplication to multilingual text, with controlled pipeline ablations on nine languages before broader deployment [27]. Language-aware processing matters because a rule expressed in whitespace-delimited words does not have the same meaning across writing systems. The Common Pile instead organizes collection around public-domain and openly licensed sources and releases processing code, training mixtures, and model checkpoints [28]. Its 8 TB collection and the 1T- and 2T-token training runs of the Comma models illustrate why stored volume and exposure should remain separate.

Acquisition itself is also becoming selective. Data Scout expands a topic into search queries, groups retrieved seeds by subdomain, and uses a small classifier-screened sample to decide where to crawl more deeply [29]. Host-aware crawling similarly allocates crawl depth using quality estimates from sampled pages [30]. The latter measures the yield of classifier-selected pages in crawl simulations, so downstream performance gains still require training experiments. These approaches move selection upstream from an existing corpus $\mathcal{D}$ to the acquisition of $\mathcal{R}$, where fetch cost, coverage, and the bias of the screening classifier jointly determine what becomes available for training.

Domain sources complement web text with code repositories, scientific papers, books, and encyclopedias. These sources contribute structures and content that generic web samples may underrepresent, although their value still depends on quality and provenance. Open pipelines support reproducible extraction, normalization, deduplication, and quality control for domain strata $\mathcal{D}_k$.

PDFs and scans provide another route to scientific and long-form text, but their layout must be converted into reliable sequences. Formulas, tables, multiple columns, and reading order are recurrent extraction difficulties. Nougat [33] uses an end-to-end Transformer to transcribe academic documents, including mathematical structure. MinerU [34] addresses a

Table II. Major open pretraining corpora. Performance numbers are color-coded by evaluation setup: **FW-Eval** = aggregate of 8 NLP benchmarks, 1.8B model, 350B tokens [14]. **DCLM-Eval** = MMLU, 7B model, 280B tokens [15]. **NC-Eval** = average of 20 benchmarks, 8B model [26]. Comparisons require matching protocols within each group, and scores across groups are not directly comparable. Scale denotes the reported corpus or candidate-pool size, not a common training budget. DCLM 240T denotes its candidate pool, not the filtered training set.

Corpus	Scale	Design Focus	Performance
C4 [24]	156B	Rule-filtered CommonCrawl corpus	FW-Eval 48.2
The Pile [22]	825 GiB	Add data sources to broaden general knowledge	FW-Eval 46.8
RefinedWeb [13]	5T	Web-only data can match curated multi-source corpora	FW-Eval 48.8
SlimPajama [31]	627B	Global deduplication across sources	FW-Eval 46.0
Dolma [25]	3T	Fully open and reproducible pipeline	FW-Eval 47.5
RedPajama V2 [32]	30T	Flexible quality filtering with 40+ quality signals	FW-Eval 46.7
FineWeb [14]	15T	Stage-by-stage ablations of the data pipeline	FW-Eval 48.8
FineWeb-Edu [14]	1.3T	Filtering targeted at educational content	FW-Eval **50.5**, DCLM-Eval 37.3, NC-Eval 53.2
DCLM [15]	240T	Standardized data-selection benchmark	DCLM-Eval **50.8**, NC-Eval 57.0
Nemotron-CC [26]	6.5T	Classifier ensembling and synthetic rewriting	NC-Eval **57.8**

wider range of digital documents and scans, while olmOCR [35] uses a vision-language model for page-level transcription. OLMo 3 [36] incorporates extracted academic and book content and uses synthetic question–answer data in mid-training. Transcription fidelity and source quality both matter.

Learned extraction increasingly combines scraping and cleaning in one process. ReScraper trains a compact model to extract content and then retain, edit, discard, or rewrite each page [37]. Comparisons on the same crawled pool use 400M–2.8B models. Provenance records should therefore distinguish extraction from generation.

B. Deduplication

Deduplication controls accidental repetition in the stored corpus. Web redundancy includes identical or near-identical content across URLs and repeated observations of the same page across crawl snapshots. Unlike deliberate resampling, deduplication changes $\mathcal{D}$ and its implicit source weights. Resampling changes exposure q_s while preserving the corpus.

Exact deduplication compares normalized content hashes, whereas URL matching identifies repeated addresses and can miss changes to their content. OLMo 1 [38] and the Dolma pipeline [25] use Bloom-filter-based processing to reduce memory use during duplicate detection. The false-positive rate should be reported because it affects document retention.

Near-duplicate detection also captures documents that share substantial text without matching exactly. MinHash estimates Jaccard similarity, the fraction of distinct token n-grams shared by two documents relative to their union. Locality-sensitive hashing (LSH) retrieves likely matches without comparing every document pair. These methods measure lexical overlap rather than semantic equivalence. Similarity thresholds, clustering rules, and the choice of retained representatives determine which documents survive. Multiple signatures improve estimation, while LSH trades candidate recall against comparison cost. With a fixed signature count, storage grows approximately linearly with corpus size, although efficiency depends on bucket skew and the handling of short documents.

Reported pipelines operate at different granularities and cover different sources. LLaMA 3 combines URL, document-level, and line-level deduplication to address both replicated documents and recurring fragments. DeepSeek V2 [39] describes cross-snapshot processing to control temporal redundancy, which remains relevant to later releases. InternLM 2 [40] reports MinHash settings including 128 hash functions, 5-grams, and a Jaccard threshold of 0.7. It also describes code-specific organization to preserve useful repository structure. These settings make the procedure reproducible, but they require validation before reuse in another corpus. Cross-source and cross-snapshot coverage can matter as much as the matching algorithm.

Aggressive removal can reduce useful coverage when independently written documents contain shared explanations or common facts. Overlap need not imply equal training value, while frequency can reflect common knowledge, scraping artifacts, or spam. Choosing thresholds and cluster representatives therefore requires checking which domains and useful examples are retained.

Empirical results on repetition depend on the source of redundancy and the data budget. Hernandez et al. [41] show that concentrated repetition can reduce effective model capacity in their scaling experiments. Carlini et al. [42] associate repeated training sequences with increased verbatim memorization. These findings support controlling unintended repetition, but do not imply that every additional epoch is harmful. Muennighoff et al. [43] show that repeated training can remain beneficial when fresh data is limited, with diminishing returns. Repetition should therefore be compared with alternative uses of the same budget B, including fresh text.

Recent domain-level experiments refine this distinction fur-

ther. Under a fixed tokens-per-parameter ratio, Scaling Domain Data Repetition finds that the preferred repetition count can increase mildly with model size and varies across domains [44]. The study relates those differences more strongly to final domain validation loss than to unique domain size. Its setting concerns deliberate reuse of a domain within a mixture, rather than accidental copies in a crawl. A deduplicated corpus can therefore support multiple epochs, with reuse compared against other allocations of the same budget.

C. Quality Filtering

Quality filtering selects documents according to an operational criterion. Let $s_\phi(x)$ be a scoring function and τ_q a threshold. A deterministic filter produces

$$\mathcal{D}_{\text{keep}} = \{x \in \mathcal{D} : s_\phi(x) \geq \tau_q\}. \tag{3}$$

Soft selection instead modifies the probability of sampling a document. Neither formulation supplies a task-independent definition of quality: the labels, reference distribution, and downstream evaluation determine what the score rewards. Aggregate benchmark gains should therefore be reported alongside retention rates and domain composition.

Practical pipelines often combine inexpensive rules with learned scoring. The Gopher rules [45] threshold document length, word statistics, and repetition-related features to remove obvious extraction artifacts and low-information text. Perplexity-based filtering measures how well a reference language model predicts each document. Yi [46] and Falcon [47] report such filtering, although a high perplexity may indicate domain novelty as well as corruption. Rules and reference-model scores are consequently useful as coarse signals whose thresholds require validation.

Learned filters depend strongly on how positive examples are defined. GPT-3 uses reference sources including Reddit-linked content, and LLaMA 1 [21] uses Wikipedia-related quality signals in its CommonCrawl processing. LLaMA 3 distills LLaMA 2-generated quality judgments into a smaller classifier. DCLM [15] studies fastText-based selection with curated positive data, including instruction-style examples, and reports a 7B model reaching approximately 64% on MMLU. Its reported compute comparison with LLaMA 3 is an end-to-end model comparison, not an isolated causal estimate of the filter. Controlled comparisons within the DCLM framework provide the stronger basis for attributing gains to selection.

FineWeb-Edu [14] uses Llama-3-70B-Instruct to score 500K examples for educational value and trains a lightweight classifier to scale the selection procedure. Nemotron-CC [26] combines complementary quality signals and synthetic transformations, while Qwen 3 describes multidimensional annotation and data selection. These systems separate expensive judgment generation from lower-cost corpus-wide scoring. The resulting labels can inform both Eq. (3) and the domain weights used during training. Their cost depends on annotation volume, classifier complexity, and corpus size.

Recent methods assess quality along several dimensions rather than relying on a single criterion. Meta-rater combines professionalism, readability, reasoning, and cleanliness with existing scores, using proxy experiments to learn their relative weights [48]. QuaDMix jointly parameterizes quality-based selection and domain allocation, rather than treating their composition as fixed [49]. Leverage Score Sampling introduces an embedding-space diversity criterion that favors samples expanding the represented feature space [50]. These measures capture diversity or predicted quality without establishing factual correctness. Training experiments and coverage audits are needed to assess their usefulness.

Multilingual filtering introduces an additional transfer problem. Recent work adapts English quality classifiers by scoring machine-translated text and distilling those labels into a multilingual classifier [51]. Experiments at 1B, 3B, and 8B parameters examine downstream performance together with regional and cultural knowledge. This reduces the need for separate annotation in every language, while retaining dependence on translation quality, encoder coverage, and the original scoring criterion. Multilingual evaluation should therefore report per-language retention and regional coverage.

Rather than applying the same operation to every document, adaptive pipelines choose how to process each example. DataOrchestra chooses whether to drop, retain, or clean a passage and composes programmatic edits with instructed rewriting when needed [52]. AutoData instead searches executable scoring, stratification, and stochastic-selection programs over document features, using small-model validation as feedback [53]. Assessing the benefits of such adaptation requires reporting the available operations, search costs, and reuse of evaluation data. Transfer across the small models studied in AutoData remains to be validated at frontier scale.

The principal limitation is selection bias. A classifier trained on encyclopedic or educational prose may reject informal documents that still contain useful information. DeepSeek V2 reports that relaxing filters can benefit knowledge-intensive tasks, whereas Yi emphasizes a smaller, more selective corpus. Neither observation establishes a universally optimal threshold. Filter evaluation should specify the retained distribution, exposure, and target capabilities.

D. Synthetic Data Augmentation

Synthetic augmentation uses a model to generate or transform text. Let q_{nat} and q_{syn} be distributions over natural and accepted synthetic training examples. Their mixture is

$$q_\rho = (1-\rho)q_{\text{nat}} + \rho q_{\text{syn}}, \qquad 0 \leq \rho \leq 1. \tag{4}$$

Here ρ gives the probability of sampling a synthetic example, while the resulting token share also depends on example lengths. Synthetic generation may be grounded in a source document or conditioned only on a prompt and the parameters of the generator. Interpreting the mixture therefore requires the seed provenance, validation rules, and sampling convention.

Generative augmentation. Generative augmentation introduces new text whose composition depends on the generation task and model family. The description of web and ebook data in the report of DeepSeek V3 does not by itself establish that every pretraining example is free of model-generated content. Phi-4 [54], in contrast, explicitly assigns a substantial

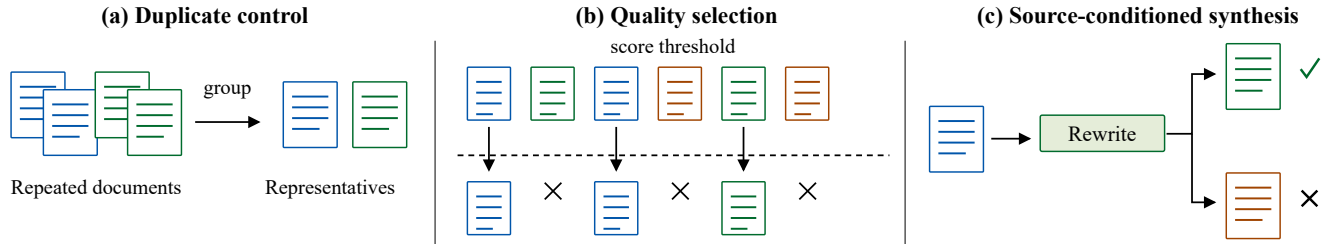


Fig. 3. Corpus construction and model-assisted processing. Filtering selects existing examples, whereas synthesis changes their content. Both affect the retained distribution and require validation before sampling for training.

role to synthetic data, reporting a 40% synthetic share in its pretraining mixture. Phi-1 [55] demonstrates the use of generated textbook-style material for code learning, and later Phi models expand the approach. Phi-4 reports more than 50 synthetic dataset categories with approximately 400B stored synthetic tokens. That stored volume and the synthetic share of training exposure are different quantities. Transfer beyond the model scales studied in Phi remains unverified.

Other models use synthesis to strengthen particular domains. Qwen 2.5 [56] employs specialized models for mathematical and code data, and Qwen 3 extends generation to a broader range of subjects and formats. Related practices in InternLM use a preceding model generation to support subsequent corpus construction. OLMo 3 [36] combines document extraction with question–answer generation for mid-training. These examples show how synthesis can supply targeted contexts and supervision while retaining a natural-text foundation. They do not establish a universal synthetic-data ratio.

Source-conditioned rewriting. A complementary approach transforms existing documents while retaining a source for validation. Kimi K2 [57] rewrites documents into different styles and perspectives, including mathematical notes and translated variants, and limits the number of rewrites per source. Its SimpleQA comparison supports rewriting over straightforward repetition in the reported setting. Nemotron-CC describes approximately 1.9T generated tokens, with transformations chosen according to source quality, including denoising, question–answer pairs, summaries, and knowledge lists. Grounding a rewrite in a source can constrain its content, but cannot guarantee factual preservation or eliminate the stylistic and distributional biases of the generator. Validation must therefore address both correctness and diversity.

Recent rewriting studies examine how these transformations depend on source and purpose. SwallowCode and SwallowMath refine existing code and mathematical text through domain-specific validation and rewriting, with evaluation by continued pretraining [58]. BeyondWeb studies source selection, rewriting style, and generator choice across several pretraining configurations [59]. Synthesis therefore involves choosing transformations and outputs to retain. Comparisons should include synthesis costs and distinguish continued training from training from scratch.

Other work changes the structure of the generated supervision. FineInstructions matches instruction templates to source documents and constructs grounded instruction–answer pairs at pretraining scale [60]. Its from-scratch experiments provide evidence about corpus transformation, although their instruction-conditioned objective differs from both all-token web pretraining and subsequent instruction tuning. Synthetic megadocs instead combine multiple rephrases of the same document, or extend it with inserted rationales, to create longer training sequences [61]. In its data-constrained experiments, sequence construction improves on treating the rephrases as independent short examples. This result connects synthesis to $\mathcal{T}_s$ and L_s, showing that the arrangement of generated text can matter in addition to the mixture coefficient ρ.

Controlled experiments further show that benefits depend on the natural–synthetic mixture. Kang et al. [62] report improvements in training efficiency for selected mixed-data settings, including a one-third synthetic mixture, while purely synthetic alternatives do not consistently outperform natural web data. The reported 5–10× gains are specific to their models, tasks, and efficiency definitions. Recursive replacement of natural data can also cause distributional degradation under the conditions studied in model-collapse research [63]. These findings motivate preserving natural coverage and validating successive synthesis rounds.

IV. Data Utilization

Data utilization determines how the corpus $\mathcal{D}$ contributes to parameter updates through the policy π_s. We review selection and mixing (§IV-A), multi-stage training (§IV-B), long-context training (§IV-C), and fine-grained training signals (§IV-D).

A. Data Selection and Mixing

For K domain strata, a mixture can be represented as

$$q_s(x) = \sum_{k=1}^{K} \alpha_{s,k} q_k(x), \quad \alpha_{s,k} \geq 0, \quad \sum_{k=1}^{K} \alpha_{s,k} = 1, \tag{5}$$

where q_k specifies sampling within stratum $\mathcal{D}_k$. If $\alpha_{s,k}$ denotes document-sampling mass and the expected transformed length in domain k is $\bar{L}_{s,k}$, the corresponding expected token share is

$$\widetilde{\alpha}_{s,k} = \frac{\alpha_{s,k} \bar{L}_{s,k}}{\sum_j \alpha_{s,j} \bar{L}_{s,j}}. \tag{6}$$

Thus document and token mixing ratios coincide only under appropriate length normalization. Reported percentages should be interpreted in their original sampling convention.

Early models commonly fixed mixture weights using source availability and development experiments. LLaMA 1 reports

a mixture including CommonCrawl at 67%, C4 at 15%, and GitHub at 4.5%. BLOOM [64], Falcon [47], and PaLM [20] also describe predetermined source or language mixtures. PaLM 2 [65] increases emphasis on multilingual and code data, whereas OLMo 1 uses the Dolma recipe without an extensive mixture search. Fixed ratios simplify reproduction but need validation under the intended training configuration.

The definition of a domain is itself part of mixture design. WebOrganizer annotates web text by topic and format, making it possible to distinguish changes in subject matter from changes in presentation [66]. Nemotron-CLIMB instead forms semantic clusters and iteratively searches their mixture with proxy training and performance prediction [67]. ToReMi uses topic assignments and observed learning dynamics to reweight examples, addressing variation within a source as well as between sources [68]. Thus the strata $\mathcal{D}_k$ need not coincide with acquisition sources. Their granularity determines which distributional changes Eq. (5) can express and how much evidence is needed to estimate useful weights.

Proxy-based optimization reduces the cost of exploring Eq. (5). A generic formulation selects

$$\alpha^* \in \arg\min_{\alpha \in \Delta^{K-1}} \widehat{\mathcal{R}}_{\mathcal{E},\Gamma}(\alpha, P, B), \tag{7}$$

where Δ^{K-1} is the probability simplex and $\widehat{\mathcal{R}}$ is a surrogate for evaluation risk on data $\mathcal{E}$ under protocol Γ. LLaMA 3 reports scaling-law experiments to refine its mixture and subsequent adjustments informed by capability measurements. Qwen 2.5, Qwen 3, and DeepSeek V3 describe related data-allocation procedures. These disclosures support the use of empirical optimization, without demonstrating adoption of a particular academic algorithm.

Proposed methods differ in how they predict mixture quality and search for useful weights. DoReMi [69] uses a proxy model and distributionally robust optimization based on domain excess loss to estimate weights. Data Mixing Laws [70] fits an analytic relationship between mixtures and loss, reporting training-step savings in its experiments. RegMix [71] learns a regression predictor from small runs and reports lower search cost than its DoReMi baseline. BiMix [72] models the joint effect of mixture and training volume. UtiliMax [73] uses utility estimates and allocation constraints to formulate mixture design as portfolio optimization. Their reported savings refer to different search and training settings, so they should not be combined into a single efficiency ranking.

Subsequent work develops alternative approximations to the mixture objective. MixMin derives a convex formulation under assumptions on the expressiveness of the model class and approximates it using source-specific predictors [74]. TANDEM uses a penalized bi-level formulation with proxy and reference networks to estimate which domains benefit from additional data [75]. Scaling Laws for Optimal Data Mixtures jointly models target loss as a function of model size, token budget, and domain proportions [76]. These methods differ in what they approximate: an idealized population objective, an optimization response, or an empirical scaling relationship.

Online methods use the current training state rather than a fixed corpus score alone. PiKE adjusts mixture weights using gradient statistics under a low-conflict multi-task formulation [77]. At the example level, OptiSelect studies how optimizer-dependent scoring geometry changes which candidates appear useful [78]. Its 124M- and 720M-parameter experiments demonstrate that a ranking based on raw gradients need not match a ranking based on the effective update. Document utility therefore depends on the training state, target distribution, and scoring method.

The usefulness of these optimization methods depends on whether proxy results remain informative at the target scale. DataDecide [79] studies more than 30K checkpoints and finds substantial, but imperfect, transfer of data rankings. AutoScale [80] identifies settings where small-scale optima do not transfer, and Mizrahi et al. [81] show that preferred filtering thresholds can vary with model size. Applying proxy results at a larger scale therefore requires sensitivity checks for model size P, budget B, and the evaluation protocol in Eq. (7).

B. Multi-stage Training

Multi-stage training changes the utilization policy over the course of optimization. A piecewise-constant data schedule can be written as

$$\pi_s = \pi^{(j)} \quad \text{for } s \in I_j, \qquad j = 1, \ldots, J, \tag{8}$$

where the intervals I_j partition the training steps into stages. Each stage may change domain weights, sequence transformations, context length, or target weights. The learning-rate schedule η_s can change independently. Separating these controls helps distinguish the effect of a data change from that of a learning-rate change.

Several reports allocate most exposure to broad pretraining before emphasizing selected capabilities. Qwen 3 describes an initial stage with more than 30T tokens spanning 119 languages and dialects, a subsequent approximately 5T-token stage emphasizing science, code, and reasoning, and a final long-context stage. GLM-5 and DeepSeek V3 also report staged training, although their boundaries and objectives differ. LLaMA 3 adjusts data allocation together with optimization settings during training. Stage counts and token allocations vary across these reports.

Mid-training usually denotes continued training between broad pretraining and later adaptation, but reports use the term inconsistently. OLMo 2 [82] and OLMo 3 [36] emphasize higher-quality mathematical, code, and question–answer data in later training. The report for InternLM 2 [40] also describes capability-oriented continued training. Our scope covers language-model training on text, excluding instruction tuning and reinforcement learning.

Annealing denotes learning-rate decay and is often accompanied by an increase in selected high-quality data. LLaMA 3 reports a 24-percentage-point GSM8K improvement for its 8B model during its final annealing phase, while the gain differs at larger scales. Related late-stage selection is described for OLMo 3 and Phi-4. A decaying learning rate, such as one reported for Kimi K2, does not by itself establish a simultaneous quality-focused data switch. MiniCPM [83] studies

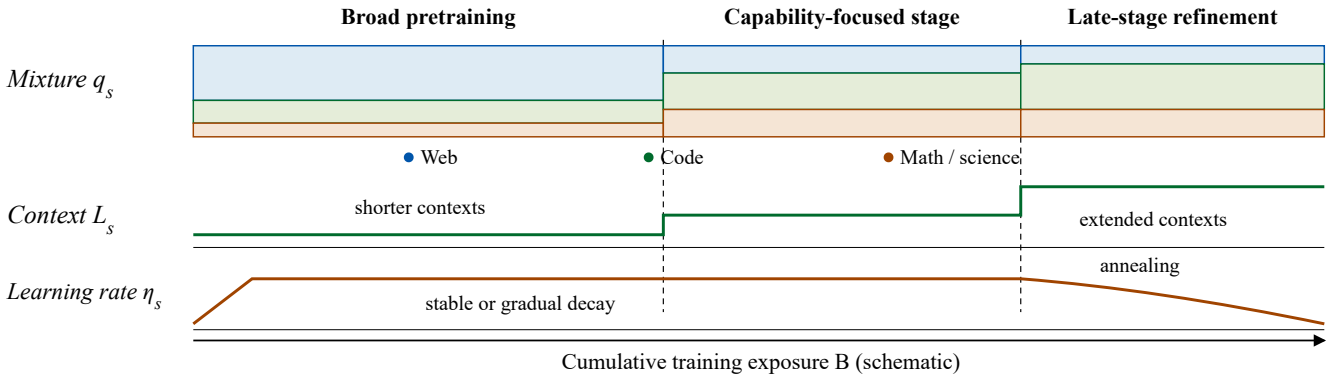


Fig. 4. A schematic staged utilization policy. The mixture, context bound, and learning rate are separate controls. Stage widths and curves are illustrative and do not represent measured token allocations or a universal recipe.

the warmup–stable–decay (WSD) schedule. Luo et al. [84] analyze interactions between learning rate and data quality, providing one account of why a late data change may have different effects from the same change earlier in training. GLM-4.5 reports limitations of WSD in its setting, so WSD should not be presented as universally preferable.

The terms therefore describe different aspects of Eq. (8): mid-training locates a stage within the training process, while annealing specifies learning-rate decay. A stage may involve both, in which case controlled comparisons are needed to separate their effects.

Classical curriculum learning orders examples by a notion of difficulty, while capability-oriented staging primarily changes domains or target skills. Zhang et al. [85] report training-step reductions of 18–45% for the curriculum settings they study. Frontier staging need not follow this difficulty ordering, whose definition across heterogeneous text remains unresolved. Fig. 4 therefore depicts policy changes without universal difficulty labels.

Recent scheduling methods replace manually fixed transitions with predictions or feedback. RegMix-D fits local loss transitions from proxy training trajectories and uses them to produce an offline mixture schedule or update the mixture from observed target-model loss [86]. The Holistic Data Scheduler learns an online allocation policy from rewards combining data quality, loss-related inter-domain effects, and model statistics [87]. The controller adjusts sampling within the existing training procedure.

Scheduling also interacts with optimization and retention. Quality-aware functional scaling laws analyze the placement of cleaner data jointly with batch size in a feature-space regression model, then motivate a schedule evaluated in language-model mid-training [88]. The theoretical optimum is conditional on that model and should not be read as a general law for all pretraining trajectories. In continued pretraining, Replay on Demand allocates exposure between new-domain samples with remaining learning potential and replay samples showing forgetting [89]. Scheduling thus balances acquiring new capabilities with retaining existing ones.

C. Long-context Training

Long-context training increases the context bound L_s and changes which dependencies can contribute to learning. It supports applications such as long-document analysis and repository-level code understanding, but a larger nominal window does not ensure effective use of distant information. LLaMA 3, DeepSeek V3, Qwen 3, and Kimi K2 describe context extension after much of their shorter-context training. Both scheduling and long-range data dependencies require empirical validation.

Staged extension limits the cost of applying long contexts throughout pretraining. DeepSeek V3 reports approximately 14.8T tokens in its initial 4K-context training and uses YaRN [90] in later extension to 32K and 128K. LLaMA 3 increases the context window from 8K to 128K through several stages. Similar phased approaches appear in Kimi K2. Positional adaptation, sequence length, and corpus composition are coupled choices, so context-length effects should be evaluated at both short and long ranges.

Long natural documents are less abundant than short web passages, and their source distribution differs. Books, papers, and code repositories supply extended structure but may concentrate exposure in a small number of domains. Upsampling these sources changes q_s, whereas packing changes $\mathcal{T}_s$. In-Context Pretraining [91] retrieves related documents for concatenation, aiming to provide more informative cross-document context. Best-fit Packing [92] addresses length utilization by reducing padding and truncation. Packing efficiency and long-context coherence are distinct objectives.

Dependency-aware selection offers a more direct criterion than document length. LADM aggregates attention-based measurements of relationships between spans to rank long examples and evaluates the selected data through continued training [93]. Synthetic megadocs provide a complementary construction strategy by linking multiple source-related generations [61]. Both approaches seek useful dependencies rather than length alone. Because attention scores depend on the model and generated links may be unreliable, evaluation should test distant evidence under matched sequence lengths.

Document boundaries require explicit attention and loss policies. For packed sequence positions with document iden-

tifiers $d(i)$, a document-isolated causal mask is

$$M_{ij} = \begin{cases} 0, & j \le i \text{ and } d(j) = d(i), \\ -\infty, & \text{otherwise.} \end{cases} \tag{9}$$

Allowing cross-document attention removes the second condition and changes the statistical training context. End-of-document tokens alone do not enforce attention isolation. Reports should also specify whether boundary predictions contribute to the loss. Synthetic retrieval or long-context examples, as described in later stages of Qwen 3 and other models, provide an additional way to create distant dependencies. Their benefit should be measured with held-out long-context tasks rather than inferred from the number of packed tokens.

D. Training Signals

Standard language modeling assigns equal target weights in Eq. (2), but token losses and gradients are not equal. Xia et al. [94] show that learning trajectories vary across tokens and model sizes. Chang et al. [95] observe learning, forgetting, and subsequent stabilization within training. These results motivate adjusting target contributions, although a frequently occurring or already predictable token is not automatically uninformative. The effect of an intervention depends on the full context and the normalization of the objective.

Selective token supervision. RHO-1 [96] selects tokens using excess loss relative to a reference model trained on high-quality data. A simplified selection rule is

$$w_{s,i,t} = \mathbf{1}[\ell_{i,t}(\theta_s) - \ell_{i,t}(\theta_{\text{ref}}) \ge \gamma_s], \tag{10}$$

where γ_s can be chosen to retain a prescribed fraction of targets. The rule prioritizes positions that remain difficult for the current model but are predictable to the reference. Reference-model quality and domain coverage therefore influence which signals are retained. This changes the optimization objective, rather than merely saving storage or removing documents.

Sequence transformation for infilling. Fill-in-the-Middle (FIM) [97] instead changes the sequence transformation. For a prefix p, middle span m, and suffix u, a representative arrangement presents p and u before the target m, separated by special markers. The model learns $p_\theta(m \mid p, u)$ by decoding the transformed sequence with causal attention in the new ordering. StarCoder [98], Code LLaMA [99], and DeepSeek-Coder-V2 [100] use this approach for code. DeepSeek V3 and DeepSeek V4 also describe infilling-related training. At the evaluated transformation rates, these studies report compatibility with left-to-right generation.

Distillation targets. Knowledge distillation changes the target distribution. With teacher probabilities p_{T} at the same prediction context, a representative token objective is

$$\ell_{i,t}^{\text{KD}} = -\sum_{v \in \mathcal{V}} p_{\text{T}}(v \mid c_{i,t}) \log p_\theta(v \mid c_{i,t}). \tag{11}$$

Gemma 2 [101] uses pretraining distillation for its 2B and 9B models. Teacher targets convey information about alternative continuations, but also incur inference costs and may transfer teacher biases. They should be distinguished from sampling teacher-generated documents, which changes the corpus itself. Their relative weights must be specified when both are used.

Metadata as conditioning context. Metadata conditioning exposes contextual information that ordinary text extraction may discard. A document's URL, source hierarchy, or timestamp can be represented as a prefix z, yielding body-token predictions $p_\theta(x_t \mid z, x_{<t})$ with the metadata positions excluded from the loss. Allen-Zhu and Li [102] motivate studying how presentation context affects knowledge acquisition. MeCo [103] reports faster convergence using metadata prefixes, including matching a standard baseline with approximately 33% fewer training tokens in its settings. Beyond URLs [104] studies the granularity of metadata, and KoCo [105] represents metadata through composable knowledge coordinates. Their benefits depend on whether the metadata is reliable and whether comparable context is available at inference time. Public evidence for transfer to frontier-scale pretraining remains limited.

V. Pretraining Evaluation

Evaluation asks whether a data decision improves model performance at an acceptable cost. We distinguish model measurement (§V-A), data-recipe comparison (§V-B), and limitations of the resulting evidence (§V-C).

A. Measuring Model Performance

A model can be evaluated through prediction loss, task performance, or a benchmark aggregate. We use a general function to express their shared dependence on the model, evaluation data, and measurement protocol.

Definition 3 (Evaluation function)**.** *An evaluation function maps a model with parameters θ, evaluation data $\mathcal{E}$, and a protocol Γ to a numerical measure of performance:*

$$A : (\theta, \mathcal{E}, \Gamma) \mapsto A(\theta, \mathcal{E}, \Gamma) \in \mathbb{R}. \tag{12}$$

The protocol specifies how inputs and model outputs are processed, how performance is scored and aggregated, and whether larger or smaller values are preferred.

This definition accommodates accuracy, held-out loss, corpus-level metrics, and composite scores without prescribing a particular aggregation rule. Several functions can be reported together when one number would obscure task differences. For the score-difference and efficiency comparisons below, we orient A so that larger values are better, negating a loss when needed. The named loss metrics retain their usual lower-is-better interpretation.

Table III compares the main measurements used in this section. Prediction metrics describe how well a model assigns probability to text, task metrics assess its answers, and resource metrics measure the cost of reaching a specified outcome. Reporting them together makes the trade-offs of a data recipe easier to interpret.

Training loss and gradient statistics provide within-run diagnostics. Spikes, plateaus, and drift can signal changes in data or numerical behavior, although they do not identify the cause. PaLM describes recovery from unstable steps, while

Table III. Comparison of pretraining evaluation metrics. Arrows indicate the preferred direction. Training diagnostics do not have a universal preferred magnitude. Fair comparisons require common evaluation data and a specified protocol.

Metric and direction	What it measures	Value for data engineering	Comparison requirements and limits
Training loss and gradient statistics	Loss, gradient norms, and changes within a training run.	Detect instability, plateaus, or shifts after data changes.	Depend on sampled data and optimization. Lower training loss alone does not establish better generalization.
Held-out NLL ↓ and perplexity ↓	Mean token prediction loss, Eq. (13), and its exponential.	Provide continuous signals for comparing checkpoints and data mixtures.	Match tokenizer, corpus, context, and loss positions. Perplexity is a monotonic transform of NLL, not independent evidence.
Bits per byte ↓	Total negative log-likelihood divided by the number of raw-text bytes, Eq. (14).	Reduces the effect of token granularity when comparing models.	Match raw text, normalization, boundaries, and context. Byte normalization alone does not align protocols.
Correct-answer NLL ↓	Negative log-probability of the reference answer under a task prompt.	Reveals gradual changes that discrete accuracy may miss and supports calibrated extrapolation [106–108].	Fix prompts, answer representation, and length normalization. Sensitivity to probability changes does not guarantee task success.
Accuracy / exact match ↑	Fraction of answers satisfying the task scoring rule.	Directly summarizes success on knowledge, reasoning, and question-answering tasks.	Fix demonstrations, decoding, answer extraction, and normalization. Discrete scores can hide gradual improvements.
Code pass@k ↑	Probability that at least one of k generated solutions passes the tests.	Measures functional correctness on code benchmarks [109–111].	Match k, sampling settings, sample count, execution environment, and test suite. Passing tests does not prove correctness for all inputs.
Benchmark aggregate ↑	A specified combination of task scores, each oriented so that larger is better.	Summarizes broad performance across a fixed task suite.	State task weights, normalization, and missing-task handling. Report individual scores to expose domain regressions.
Compute / tokens to target ↓	Compute $F_i(a_0)$ or exposure needed to reach the same target, Eq. (16).	Compares resource efficiency across data recipes.	Fix the target and protocol, report unreached targets, and distinguish training costs from processing, synthesis, and search costs.

DeepSeek V3 and Kimi K2 report monitoring and stabilization procedures. Gradient and parameter norms inform rollback or batch exclusion, but absolute losses remain incomparable across tokenizers and distributions.

Held-out negative log-likelihood provides a continuous measure of prediction quality. For fixed tokenized evaluation data, define

$$\mathcal{L}_{\mathcal{E}} = -\frac{1}{T_{\mathcal{E}}} \sum_{x \in \mathcal{E}} \sum_{t} \log p_\theta(x_t \mid x_{<t}), \tag{13}$$

where $T_{\mathcal{E}}$ counts evaluated targets. Lower loss indicates better prediction, and perplexity is $\exp(\mathcal{L}_{\mathcal{E}})$. For a raw-text evaluation corpus, bits per byte (BPB) is

$$\mathrm{BPB}(\theta, \mathcal{E}) = \frac{-\sum_{x \in \mathcal{E}} \log p_\theta(\tau(x))}{\ln(2) \sum_{x \in \mathcal{E}} |x|_{\mathrm{bytes}}}. \tag{14}$$

Byte normalization reduces dependence on tokenizer granularity, but does not align text normalization, context truncation, boundary handling, or evaluation samples. Uses in DeepSeek V3 and OLMo 3 should be interpreted with these protocol details fixed.

Downstream benchmarks assess specific tasks rather than average text prediction. Table IV records benchmark coverage, while Table III distinguishes the scoring rules applied to their outputs. Few-shot prompting and likelihood-based option scoring reduce output-format ambiguity for base models. Accuracy and exact match measure discrete success, so they may remain flat despite likelihood gains. For code generation, pass@k measures success across sampled solutions and depends on both the sampling budget and test suite.

Schaeffer et al. [106] show that continuous metrics can reveal smoother trends where discontinuous metrics suggest emergent behavior. Their findings do not imply that every transition is an artifact. Gu et al. [107] find different early-pretraining signals from answer-letter and cloze-style protocols. Comparisons require the full protocol Γ and complementary metrics such as NLL and accuracy.

B. Comparing Data Recipes

A data recipe combines corpus construction with the utilization policies applied during training. To compare two recipes, train a model with each and evaluate both on the same examples under the same protocol. Let θ_1 and θ_2 denote the resulting checkpoints. Their observed score difference is

$$\Delta A = A(\theta_1, \mathcal{E}, \Gamma) - A(\theta_2, \mathcal{E}, \Gamma). \tag{15}$$

A positive ΔA means that the first model scores higher. Attributing this difference to data requires control of other experimental differences. Fig. 5 separates such comparisons from development feedback and final assessment.

Each comparison should identify the changed data decision and control architecture, optimization, budget, and checkpoint

Table IV. Benchmark coverage recorded for the representative model reports. A check denotes a recorded evaluation, while a blank indicates that the inventory contains no entry rather than confirming non-use. Coverage alone does not establish that checkpoints, prompts, or training stages are comparable.

Category	Benchmark	DeepSeek V4	LLaMA 3/3.1	Qwen 3	Kimi K2	GLM-5	InternLM 2	Gemma 3	Phi-4	Falcon 3	OLMo 3
Knowledge & Exam	MMLU [112]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	MMLU-Pro [113]	✓	✓	✓	✓			✓	✓	✓	
	C-Eval [114]	✓				✓	✓				
	AGIEval [115]	✓					✓	✓			
Reasoning & Common Sense	HellaSwag [116]	✓	✓			✓		✓			✓
	ARC-Challenge [117]		✓					✓		✓	✓
	WinoGrande [118]	✓	✓					✓		✓	✓
	BBH [119]	✓	✓	✓	✓	✓		✓		✓	
	GPQA (Diamond) [120]		✓	✓	✓			✓	✓	✓	
	PIQA [121]							✓		✓	✓
Mathematics	GSM8K [122]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	MATH [123]	✓	✓	✓	✓		✓	✓	✓	✓	
	MGSM [124]	✓		✓				✓			
Code Generation	HumanEval [109]	✓	✓		✓	✓	✓	✓	✓		
	MBPP [110]		✓		✓		✓	✓	✓		
	EvalPlus [111]		✓	✓							
Reading & QA	TriviaQA [125]	✓					✓	✓			✓
	NaturalQuestions [126]							✓			✓
	DROP [127]	✓						✓			✓

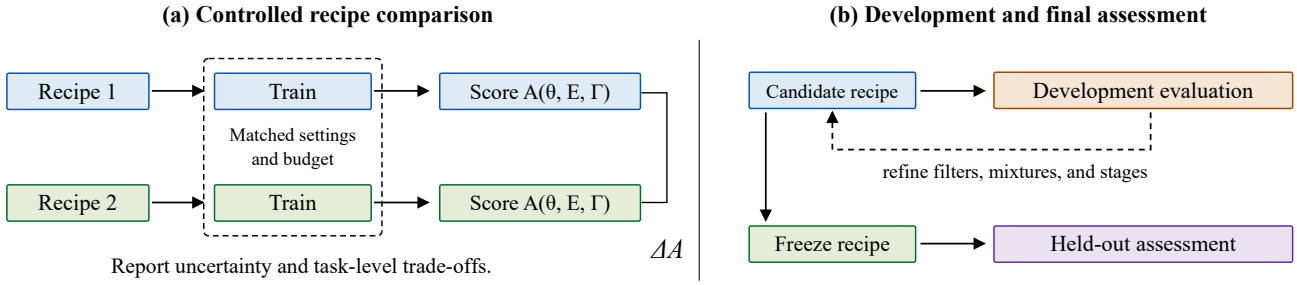


Fig. 5. Evaluation of a data intervention. Controlled training and a fixed evaluation protocol connect recipe changes to observed score differences. Development feedback is separated from held-out assessment to limit adaptive overfitting.

selection. Equal token exposure compares performance after consuming the same number of tokens, whereas equal compute fixes the training cost. Repeated runs estimate variation from initialization and sampling order, whereas evaluation-example resampling estimates uncertainty from the evaluation sample.

For a fixed target score a_0, compute efficiency can be expressed as the training compute needed to reach that target. If $\theta_{i,F}$ is the checkpoint obtained with recipe i after compute F, define

$$F_i(a_0) = \inf\{F : A(\theta_{i,F}, \mathcal{E}, \Gamma) \geq a_0\}. \tag{16}$$

The ratio $F_2(a_0)/F_1(a_0)$ measures the compute saving of recipe 1 relative to recipe 2 when both reach the target. Replacing compute F with exposure B measures data efficiency. Estimates depend on checkpoint coverage, and unreached targets should be reported. End-to-end costs also include processing, teacher inference, synthesis, and recipe search.

Checkpoint evaluation guides mixture and stage decisions. Phi-4 uses capability probes for data allocation and emphasizes retaining natural web data for knowledge tasks. Gemma 3 [128] combines benchmarks with validation perplexity. Task-level scores reveal domain regressions that an aggregate may conceal. Development tasks guide recipe selection, while held-out tasks assess the chosen configuration.

Evaluation also informs stage termination. During annealing, LLaMA 3 reports gains of 24.0 and 6.4 percentage points on GSM8K and MATH for the 8B model, with smaller gains for the 405B model. Limited gains do not establish that useful corpus information is exhausted. OLMo 2 separates development from held-out tasks, and OLMo 3 expands coverage. Deciding when to stop therefore requires weighing further gains against cost and uncertainty.

Gadre et al. [108] use correct-answer NLL to improve extrapolation from smaller runs. Their procedure first models the dependence of a continuous loss on training scale, then fits its relationship to downstream error. A representative fitted form discussed in this line of work is

$$\mathrm{Err}(L) = \varepsilon - k\exp(-\gamma L), \tag{17}$$

where parameters and validity depend on the task and calibration runs. The reported predictions, including the approximately 1/300 compute comparison, depend on the full procedure. Equation (17) is empirical and does not guarantee extrapolation beyond its fitted range. LLaMA 3 likewise uses predictive experiments to guide training.

Attribution methods provide a complementary view of which training examples contribute to an outcome. GRASP predicts the utility of a selected subset using residual alignment and a geometric interaction term, evaluated against subset-retraining results [129]. A separate trajectory-based method measures how an example update moves parameters toward the final checkpoint of a given run [130]. Such alignment need not improve every task, so training interventions are needed to validate both surrogates.

C. Measurement Challenges

Benchmark scores provide partial evidence about capability. A gain may reflect knowledge, format adaptation, contamination, or domain trade-offs. Controlled comparisons and complementary measurements help distinguish these explanations, but do not establish a causal mechanism by themselves.

Base-model evaluation may not predict subsequent adaptation. Agentic applications involve tool use, planning, and feedback absent from standard text benchmarks. Yi-Lightning [131] reports that preference rankings can diverge from benchmark performance after post-training. These observations concern evaluation transfer rather than pretraining efficacy. Ministral 3 [132] also finds that teacher rankings during pretraining distillation can differ from those after post-training.

Capability confounding further limits interpretation. MMLU combines knowledge, contextual interpretation, and answer-format compliance, which an aggregate score cannot disentangle. FineWeb-Edu reports gains on some educational tasks alongside weaker commonsense results. Task scores and domain retention statistics reveal these trade-offs.

Cross-experiment comparability adds further uncertainty. Lourie et al. [133] find that only a subset of the downstream tasks examined in the scaling setup of Gadre et al. [108] display monotonic trends. Prompt format, demonstration count, and the validation distribution can change both trends and data rankings. FineWeb-Edu and DCLM also use different model scales, training budgets, and evaluation protocols in their headline results. Those figures cannot establish a common ranking of corpora without an aligned experiment. The color groups in Table II preserve the distinction between evaluation settings rather than combining their scores.

Proxy reliability also depends on what comparison is intended. Rethinking Proxy-Model Practice shows that small changes in training hyperparameters can reverse data-recipe rankings and studies reduced-learning-rate proxies as a more reliable predictor in its experiments [134]. Holding the configuration fixed measures the effect of replacing data under those conditions. Tuning each recipe instead estimates the performance attainable within a specified search budget. These different objectives explain why joint tuning can change rankings obtained from controlled ablations.

Aggregate scores can further conceal gaps in the distribution being evaluated. FineWeb-CLaR places corpus documents and cultural benchmarks on shared culture, language, and region axes [135]. Its URL-derived regional associations provide scalable auditing metadata, not verified author locations or cultural identities. Such annotations help identify whether a filtering or mixing change removes coverage that the benchmark suite also fails to measure. Unresolved or weakly assigned metadata should remain visible in the audit rather than being interpreted as evidence of absent content.

Evaluation contamination and adaptive selection create separate risks. Text overlap between $\mathcal{D}$ and $\mathcal{E}$ may expose benchmark material during training, while repeated optimization on a development benchmark can overfit its preferences even without direct overlap. Exact and fuzzy decontamination should therefore be documented independently of ordinary corpus deduplication. Held-out evaluation should remain separate from the feedback used to choose filters, mixtures, and stage transitions. These controls support interpretation of ΔA, but a lack of detected lexical overlap does not prove that all benchmark-related information was excluded.

Recent contamination work separates detection from mitigation. DyePack embeds randomized diagnostic patterns in released benchmark material so that later activation can flag exposure under a stated statistical testing protocol [136]. It is a prospective benchmark-design method, not a retrospective detector for arbitrary unmarked tests. A controlled examination of benchmark updates finds that paraphrasing or regenerating questions does not consistently preserve measurement fidelity while removing contamination effects across the evaluated settings [137]. Those experiments use controlled contamination procedures, including fine-tuning, and therefore inform evaluation methodology without directly measuring the prevalence of contamination in frontier pretraining.

Unknown synthetic provenance is a different corpus-auditing problem from benchmark leakage. SynthSentry combines lexical diversity, n-gram tail behavior, and cross-model perplexity variation to screen batches of text for synthetic-origin signals [138]. Its evidence is limited to small-scale English experiments, and its downstream check does not establish a contamination-induced accuracy deficit that pruning reverses. The method is therefore relevant as a preliminary provenance signal rather than a validated remedy for model collapse. This distinction is especially important when a pipeline intentionally includes high-quality synthetic examples: inferred origin alone is not a measure of training utility.

Controlled synthetic evaluations can help isolate mechanisms that aggregate benchmarks confound. The Physics of Language Models studies examine knowledge storage and extraction [102], knowledge manipulation [139], and the relation between parameter count and knowledge capacity [140]. Controlled generation helps isolate causes within these experiments, although transfer to heterogeneous frontier pretraining still requires empirical validation.

VI. Trends in Pretraining Data Engineering

Across the preceding sections, corpus content, training exposure, and evaluation emerge as interdependent parts of data engineering. This section relates those decisions through a shared pipeline (§VI-A) and synthesizes the main directions emerging from model reports and controlled studies (§VI-B).

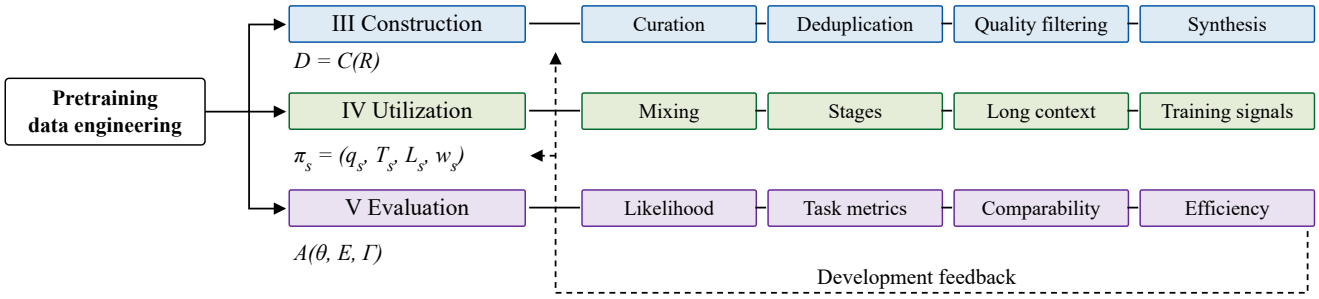


Fig. 6. Shared organization of pretraining data engineering. The pipeline follows Sections III–V, with feedback from development evaluations rather than held-out test answers.

A. A Shared Pipeline for Pretraining Data

Fig. 6 relates the three stages reviewed in this survey. Corpus construction determines the eligible examples $\mathcal{D} = \mathcal{C}(\mathcal{R})$. Utilization determines their presentation through $\pi_s = (q_s, \mathcal{T}_s, L_s, w_s)$ and interacts with the optimization schedule. Evaluation measures the resulting model through $A(\theta, \mathcal{E}, \Gamma)$, and controlled comparisons guide subsequent data decisions. Different algorithms can serve each stage, with their benefits assessed empirically.

Table V summarizes the model inventory using conservative reporting codes. An unknown entry is not evidence of absence, and synthetic post-training or multimodal processing is not automatically counted as synthetic text pretraining. A learning-rate decay alone is not coded as quality-focused annealing. Token budgets retain the reviewed sources' reporting scope: some describe repeated exposure or multimodal training, so the table should not be read as a ranking of unique text supply. In particular, Kimi K2.5 and LLaMA 4 provide broader model context rather than matched text-only experiments. Appendix A documents field-level evidence and additional model reports. Sources used only in that inventory are cited there. Read across a row, the table connects construction choices with training strategies. Its columns reveal recurring practices and gaps for further controlled study.

B. Emerging Consensus and Research Trends

The reviewed literature increasingly treats pretraining data engineering as a sequence of coupled decisions about content, exposure, and measurement. Together, these studies show why token volume alone cannot characterize either the information available to a model or how efficiently it learns. Effective recipes combine scale, processing, and allocation to meet resource constraints and capability targets.

Scale and data efficiency are complementary. Compute-optimal scaling describes one resource allocation problem [2], whereas deployment-oriented analyses introduce constraints that can favor longer training [108, 148]. RefinedWeb, FineWeb-Edu, and DCLM demonstrate that corpus processing changes the return from a fixed budget. More recent quality–diversity methods show why maximizing a document score can also narrow the retained distribution [49, 50]. The objective is therefore to achieve a desired capability profile through an appropriate combination of unique text U, exposure B, and compute F. Deduplication, filtering, and sampling contribute jointly: removing accidental copies remains compatible with deliberate reuse of scarce domain data [44].

Synthetic data extends this approach by changing the presentation of training information. Textbook-oriented generation in Phi, specialized generation in Qwen, and source-conditioned transformations in Kimi address different deficiencies in natural corpora. Recent studies connect rewriting quality to source selection, domain-specific validation, and example structure [58–61]. Across these approaches, generation aims to make existing information more useful for learning. The benefits nevertheless depend on fidelity, diversity, and the natural–synthetic mixture. Model-collapse studies identify degradation under specific recursive generation and replacement conditions [63, 149]. Their findings support preserving natural coverage and validating successive generations, rather than treating synthetic text as universally harmful. Equally, successful mixed-data recipes do not establish a universal ratio ρ. Provenance, acceptance criteria, and exposure shares remain essential to interpreting synthesis results.

Data allocation is increasingly adjusted throughout training rather than fixed in a single mixture. The fixed source proportions of LLaMA 1, iterative refinement and annealing of LLaMA 3, and distinct stages of Qwen 3 illustrate this progression. DoReMi, RegMix, and Data Mixing Laws formalize parts of this allocation problem [69–71]. Building on these ideas, subsequent methods adjust exposure using proxy trajectories, online feedback, and forgetting signals [86, 87, 89]. Such scheduling changes π_s according to capabilities still being learned or maintained, without necessarily imposing a global difficulty ordering. Transfer remains a central challenge: AutoScale documents failures of small-scale optima to transfer [80], while proxy and quality-scheduling studies connect preferred recipes to optimization conditions [88, 134]. These findings motivate intermediate-scale validation and separate reporting of data-policy changes and learning-rate transitions.

Reproducibility provides the basis for turning these observations into cumulative knowledge. The reviewed sources for LLaMA 4, Mistral 7B, and Mixtral do not disclose enough detail to reconstruct complete data pipelines [150–152], while Kimi K2 describes synthesis without a complete corpus distribution. The Common Pile, FineWeb2, and SmolLM2 offer complementary data, processing, or training artifacts that

Table V. Reported practices in the model inventory. ✓ indicates an explicit description, △ partial or indirect evidence, and ? insufficient pretraining-specific detail. Doc/Span denotes document/subdocument deduplication, Rule/Model filtering, Syn. synthetic text, Mix optimized mixtures, Stages staged training, and Q-Ann. quality-focused annealing. Model citations identify the source reports, with field-level evidence in Appendix A. These codes describe reporting, not causal effectiveness.

			Dedup		Filter			Strategy		
Model	**Date**	**Tokens**	Doc	Span	Rule	Model	**Syn.**	Mix	Stages	Q-Ann.
GLM-130B [141]	2022.10	0.4T	?	?	✓	?	?	?	?	?
BLOOM [64]	2022.07	0.35T	✓	?	✓	?	?	?	?	?
LLaMA [21]	2023.02	1.4T	✓	✓	✓	✓	?	?	?	?
Falcon [47]	2023.05	1.5T	✓	△	✓	✓	?	?	?	?
LLaMA 2 [142]	2023.07	2T	△	?	△	?	?	?	?	?
Qwen [143]	2023.09	3T	✓	?	✓	✓	?	?	?	?
Phi-2 [144]	2023.12	1.4T	?	?	△	△	✓	?	?	?
MiniCPM [83]	2024.04	1T	△	?	△	△	?	?	✓	✓
DeepSeek V2 [39]	2024.05	8.1T	✓	?	✓	✓	?	?	△	?
Gemma 2 [101]	2024.06	13T	✓	?	✓	✓	?	△	?	?
LLaMA 3 [5]	2024.07	15T+	✓	✓	✓	✓	△	✓	✓	✓
Qwen 2.5 [56]	2024.09	18T	✓	?	✓	✓	✓	✓	✓	△
DeepSeek V3 [4]	2024.12	14.8T	✓	?	✓	✓	?	✓	✓	?
Phi-4 [54]	2024.12	10T	△	?	△	△	✓	?	✓	✓
InternLM 3	2025.01	4T	△	?	△	△	△	△	△	△
OLMo 2 [82]	2025.01	4T	✓	?	✓	✓	△	△	✓	✓
Gemma 3 [128]	2025.03	14T	△	?	△	△	?	?	△	?
LLaMA 4 [145]	2025.04	40T	?	?	△	?	△	?	△	△
Qwen 3 [6]	2025.05	36T	?	?	△	✓	✓	✓	✓	△
OLMo 3 [36]	2025	5.9T	✓	?	✓	✓	✓	✓	✓	✓
Kimi K2 [57]	2025.07	15.5T	△	?	△	✓	✓	△	✓	?
GLM-4.5 [146]	2025.07	23T	△	?	△	✓	△	△	✓	?
Kimi K2.5 [147]	2026.02	15T	?	?	△	?	?	△	△	?
GLM-5 [7]	2026.02	28.5T	△	?	△	✓	△	△	✓	?
DeepSeek V4 [8]	2026.04	32T	△	?	△	△	?	△	✓	?

enable closer examination of individual choices [27, 28, 153]. Weight availability alone is insufficient: source provenance, retained distributions, processing configurations, stage-specific exposure, and evaluation protocols all affect reproducibility. Table V records reported practices, whereas controlled studies assess particular interventions. Read together, these sources help establish which findings generalize and where several effective recipes remain possible.

VII. Conclusion and Future Directions

This survey examines data engineering for text-based LLM pretraining through three connected stages: corpus construction, data utilization, and evaluation. Drawing on public model reports, open corpus projects, and controlled studies, it reviews how acquisition, deduplication, filtering, synthesis, and training schedules shape the information available to a model. A common formulation connects the constructed corpus $\mathcal{D}$, its use through policy π_s, and the resulting model performance, while distinguishing stored volume from training exposure. The comparison reveals recurring practices alongside trade-offs in coverage, quality, and computational cost. By separating reported adoption from experimentally established benefits, the survey provides a basis for interpreting current recipes and designing informative data comparisons.

Future research should strengthen the connection between data decisions and the capabilities they produce. Promising directions include validating proxy recommendations across model scales, adapting mixtures to learning and forgetting, and evaluating synthetic data for both factual fidelity and coverage. Preserving knowledge context offers another opportunity: provenance, document structure, and metadata may help models interpret information that ordinary preprocessing removes. Our ongoing work on knowledge context[1] investigates this direction, with open questions about useful representations and transfer across domains. Progress will also require evaluation that exposes domain regressions and distinguishes development feedback from held-out assessment. Releasing corpus statistics, processing configurations, stage-specific mixtures, and evaluation protocols would enable these ideas to be tested under comparable budgets and turn individual training results into cumulative evidence.

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APPENDIX A
PRETRAINING DATA DISCLOSURE EVIDENCE

Coding scheme. L1 indicates that a practice is mentioned, while L2 provides a qualitative description of sources or methods. L3 adds quantities such as token counts, proportions, filtering rates, or stage sizes. L4 includes released data, code, or recipes, or enough detail for approximate reproduction. Blank cells in the source survey are shown as –. Evidence snippets intentionally stay close to the cited technical report or model card.

Scope and interpretation. This appendix presents an inventory of reports, release notes, and model cards. Some entries concern multimodal training, distillation, supervised fine-tuning, or reinforcement learning. Such snippets remain contextual disclosure evidence and are not counted as confirmation of text-only pretraining practices in Section VI. In particular, a field headed "Mid" may reproduce a source's broader terminology rather than the pretraining definition used in the main text. "Syn." includes heterogeneous synthetic or distilled signals and does not always mean generated pretraining documents. A dash denotes missing or insufficient evidence, not confirmed non-use. Disclosure levels describe the specificity of reporting, not methodological effectiveness or independent verification. The same report may supply evidence for multiple model versions.

Inventory summary. The table contains 54 model entries. Its recorded peak codes comprise 5 at L4, 32 at L3, 11 at L2, 1 at L1, and 5 without an assigned level. These are inventory counts, not adoption rates for text pretraining. Source locators are retained for follow-up verification, including entries whose available bibliographic record is incomplete.

Table VI. Evidence table for public disclosure of pretraining data in LLM technical reports and model cards. Each cell records the highest available evidence fragment for the corresponding data-engineering aspect.

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
BLOOM 2022-07	BigScience Multilingual	L4	L4 ROOTS, 46 natural + 13 programming languages (Sec. 3.1/T1)	L4 ROOTS pipeline, tools released (Sec. 3.1/F2)	Mix: L4 language/source bytes in T1 (Sec. 3.1/T1)	Eval: L4 eval code + xP3/ ROOTS docs (Sec. 5)	[64]
GLM-130B 2022-10	Tsinghua / Zhipu Bilingual	L3	L3 Pile 1.2T, WuDao 1.0T, corpora total 400B tokens (Sec. 3.1)	L1 filtering pipeline named (Sec. 3.1)	Mix: L3 source token counts (Sec. 3.1)	Syn.: L3 MIP 5% tokens, T0++/DeepStruct (Sec. 3.1)	[141]
LLaMA 1 2023-02	Meta General	L3	L3 CC/C4/GitHub/Wikipedia/books/ arXiv/StackExchange (Sec. 2.2/T1)	L3 CCNet, Wikipedia classifier, near-dedup (Sec. 2.2)	Mix: L3 CC 67%, C4 15%, GitHub 4.5%, Wikipedia 4.5%, Books 4.5%, arXiv 2.5%, StackExchange 2% (T1)	–	[21]
ChatGLM-6B 2023-03	Tsinghua / Zhipu Bilingual chat	L3	L3 ∼1T Chinese-English tokens (README/model card)	–	Mix: L1 Chinese + English corpus	–	[154]
Pythia 2023-04	EleutherAI Research / general	L4	L4 The Pile + Pile-dedup (Sec. 3)	L4 data order + dataloaders reconstructable (Sec. 3)	Mix: L4 fixed order, all checkpoints released (Sec. 3)	Eval: L4 training/eval code released (Sec. 3/Sec. 4)	[155]
Falcon / RefinedWeb 2023-05	TII English / general	L4	L3 CommonCrawl-derived RefinedWeb, 5T web tokens (Sec. 3)	L4 MDR, URL filter, fastText, MinHash + suffix-array dedup (Sec. 3/T2/F2)	Mix: L2 web-only design (Sec. 3)	–	[13]
Baichuan 1 2023-06	Baichuan Chinese-English	L3	L3 7B 1.2T, 13B 1.4T tokens (report/ model card)	L1 cleaning/filtering named	Mix: L1 Chinese + English	–	[156]
ChatGLM2-6B 2023-06	Tsinghua / Zhipu Bilingual chat	L3	L3 1.4T bilingual tokens (README/ model card)	–	Mix: L1 bilingual corpus	–	[157]
Phi-1 2023-06	Microsoft Code	L3	L3 textbook-quality code corpus, token counts (Sec. 2)	L2 textbook-quality filtering (Sec. 2)	Mix: L3 token-scale mixture (Sec. 2)	Syn.: L3 synthetic textbooks/ exercises (Sec. 2)	[55]
LLaMA 2 2023-07	Meta General	L3	L3 public data, 2T tokens (Sec. 2.2)	L1 safety filters, personal-data filtering (Sec. 2.2/Sec. 4)	Mix: L1 factual-source upsampling (Sec. 2.2)	–	[142]
Baichuan 2 2023-09	Baichuan Multilingual / Chinese-English	L3	L3 2.6T tokens (Abstract/Sec. 1)	L1 safety/filtering named (Sec. 6)	Mix: L1 multilingual + law/medical focus (Sec. 1)	–	[158]
InternLM 1 2023-09	Shanghai AI Lab Bilingual / general	L3	L3 1.6T tokens, web/books/code/ academic/encyclopedia (Sec. 2.1/T1)	L3 language ID, rules, MinHash, safety (Sec. 2.1)	Mix: L2 multi-stage data recipe (Sec. 2.1) Mid: L2 capability-stage data (Sec. 2.1)	–	[159]
Phi-1.5 2023-09	Microsoft Reasoning / small LM	L3	L3 phi-1 corpus + synthetic textbooks (Sec. 2)	L2 quality-first corpus (Sec. 2)	Mix: L3 7B code + 20B synthetic (Sec. 2)	Syn.: L3 ∼20B synthetic textbook-like tokens (Sec. 2)	[160]

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Table VI. Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
Qwen 1 2023-09	Alibaba Bilingual / multilingual	L3	L3 ∼3T tokens, text + code (Sec. 2.1)	L3 rules, ML scoring, exact + MinHash dedup (Sec. 2.1)	Mix: L2 source/category mix (Sec. 2.1)	–	[143]
ChatGLM3-6B 2023-10	Tsinghua / Zhipu Bilingual chat / tools	L1	L1 third-generation pre-trained dialogue model (README/model card)	–	Mid: L1 tools/function-calling training data	–	[161]
Mistral 7B 2023-10	Mistral AI General	–	–	–	–	–	[151]
Grok-1 2023-11	xAI General	–	–	–	–	–	[162]
Mixtral 8x7B 2023-12	Mistral AI General / MoE	–	–	–	–	–	[152]
Phi-2 2023-12	Microsoft Reasoning / small LM	L3	L3 synthetic + filtered web data, token scale (model card)	L2 quality filters (model card)	Mix: L2 curated blend (model card)	Syn.: L3 synthetic data central (model card)	[144]
Gemma 1 2024-02	Google English / code	L3	L3 2T/6T tokens, web/code/math (Sec. 2)	L2 unsafe/PII/recitation filters (Sec. 4/Sec. 8)	Mix: L1 web + code + math categories (Sec. 2)	Syn.: L1 post-train synthetic prompts/responses (Sec. 4)	[163]
OLMo 1 2024-02	AI2 General / open science	L4	L4 Dolma, CC/Stack/C4/Reddit/ peS2o/books/Wikipedia (Sec. 2.2/T3)	L4 data + curation tools + logs released (Sec. 2.2/Sec. 5)	Mix: L4 Dolma source tokens in T3 (Sec. 2.2/T3)	Eval: L4 OLMo-Eval + Catwalk (Sec. 2.4/Sec. 5)	[38]
DBRX 2024-03	Databricks General / MoE	L2	L2 12T tokens, web/code/math/NL (report/blog)	L1 curated data pipeline	Mix: L2 curriculum/mix categories Mid: L1 continued training stage	Syn.: L2 synthetic post-training/instruction data	[164]
InternLM 2 2024-03	Shanghai AI Lab Bilingual / general	L3	L3 ∼2T+ tokens, web/code/books/ knowledge (Sec. 2.1/T2)	L3 quality/safety/dedup pipeline (Sec. 2.1)	Mix: L2 multi-stage curriculum (Sec. 2.1) Mid: L2 capability + long-context stages (Sec. 2)	Syn.: L2 knowledge/ capability augmentation (Sec. 2.1)	[40]
Yi / Yi-1.5 2024-03	01.AI Bilingual	L3	L3 3.1T English-Chinese tokens (Abstract/Sec. 2)	L3 heuristic + learned + cluster filters, MinHash (Sec. 2.1)	Mix: L3 bilingual/source mixture figures (Sec. 2.1/F2) Mid: L2 10B long-context continuation (Sec. 2.3)	–	[46]
Command R / R+ 2024-04	Cohere Multilingual / RAG	L2	L1 multilingual + retrieval data (model card)	L1 filtering named	Mix: L2 RAG/tool-use mixture Mid: L1 continued training	Syn.: L2 tool/RAG post-training data	[165]
LLaMA 3 2024-04	Meta General / multilingual	L2	L2 15T+ tokens, 4× code, >5% non-English (blog)	L2 HTML extraction, dedup, quality classifiers (blog)	Mix: L2 scaling-law data mix (blog) Ann.: L2 annealing/cooldown data (report)	Syn.: L1 Llama 2 labels for quality classifiers (blog)	[5, 166]
MiniCPM 2024-04	ModelBest / OpenBMB Bilingual / small LM	L3	L3 CC-Chn/Dolma/C4/Pile/Stack/ StarCoder (Sec. 6.3/F11)	L3 MinHash within + across corpora (Sec. 6.3)	Mix: L3 stage data distributions (F11) Mid: L2 WSD two-stage schedule (Sec. 5/ Sec. 6.2)	Syn.: L2 UltraChat/SlimOrca/ OssInstruct + proprietary SFT (Sec. 6.3)	[83]
Phi-3 2024-04	Microsoft Small LM / multilingual	L3	L3 3.3T/4.8T tokens, web + synthetic educational data (Sec. 3)	L2 heavy filtering (Sec. 3)	Mix: L2 two-phase data blend (Sec. 3) Mid: L2 long-context continuation (Sec. 3)	Syn.: L3 synthetic data core (Sec. 3)	[167]
MAP-Neo 2024-05	MAP Bilingual / open science	L4	L4 Matrix, 4.5T tokens, corpus released (Abstract/website)	L4 cleaned corpus + cleaning pipeline released (Abstract)	Mix: L3 CC 52.55%, code 22.29%, papers/books/etc. (website) Mid: L2 two-stage pre-training (website)	–	[168]
GLM-4 2024-06	Zhipu AI Multilingual / tools	L2	L2 webpages/Wikipedia/books/code/ papers (Sec. 2)	L2 exact/fuzzy dedup, filters (Sec. 2)	Mix: L1 multilingual mixture (Sec. 2) Mid: L1 tool-capability training (Sec. 3)	Syn.: L1 reasoning/tool data (Sec. 3)	[169]
Gemma 2 2024-06	Google English / general	L3	L3 2T/8T/13T tokens, web/code/ science (Sec. 3.1)	L2 unsafe/PII/decontam/ recitation filters (Sec. 3.1)	Mix: L2 mixture chosen by ablations (Sec. 3.1)	Syn.: L1 distillation targets for 2B/9B (Sec. 3.2) Eval: L2 eval-set decontamination (Sec. 3.1)	[101]
Nemotron 4 2024-06	NVIDIA General / synthetic-data generation	L3	L3 9T tokens (Introduction)	L2 high-quality dataset referenced (Intro)	Mix: L2 post-train SFT/RL domains (Sec. 3) Mid: L2 continued training (Sec. 2.3)	Syn.: L3 post-train >98% synthetic (Abstract/Sec. 1)	[170]

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Table VI. Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
Qwen 2 2024-06	Alibaba Multilingual	L3	L3 7T tokens, 27+ languages (Sec. 2)	L2 model-assisted filtering + dedup (Sec. 2)	Mix: L2 multilingual/code data (Sec. 2) Mid: L2 long-context/capability stages (Sec. 2)	Syn.: L2 synthetic high-quality data (Sec. 2)	[171]
LLaMA 3.1 2024-07	Meta Multilingual / long context	L2	L2 Llama 3 continuation, multilingual + long-context data (Sec. 3)	L2 Llama 3 pipeline inherited (Sec. 3)	Mix: L2 long-context mix (Sec. 3) Mid: L2 128K continued pre-training (Sec. 3) Ann.: L2 cooldown data (Sec. 3)	Syn.: L1 post-training/eval synthetic data (Sec. 4)	[5]
Qwen 2.5 2024-09	Alibaba Multilingual / code / math	L3	L3 18T tokens, math/code/ multilingual expansion (Sec. 2)	L3 filtering + decontamination in subreports (Sec. 2)	Mix: L3 domain stages/ratios in subreports Mid: L3 math/code continued training Ann.: L2 annealing stage	Syn.: L3 math/code synthetic data (Sec. 2, Coder/Math reports)	[56, 172, 173]
DeepSeek V3 2024-12	DeepSeek General / MoE	L3	L3 14.8T tokens, web pages + e-books, math/code (Sec. 2.1)	L2 preprocessing inherited, sparse details (Sec. 2.1)	Mix: L2 math/code + FIM emphasis (Sec. 2.1) Mid: L2 context extension + MTP (Sec. 2)	Syn.: L1 no intentional synthetic pre-training (transparency disclosure)	[4]
Phi-4 2024-12	Microsoft Reasoning / small LM	L3	L3 web/books/code/synthetic, token/ proportion facts (Sec. 3)	L2 filtering + quality selection (Sec. 3)	Mix: L2 web/books/code/synthetic blend (Sec. 3) Mid: L2 mid-training stage (Sec. 3)	Syn.: L3 ∼40% pre-training synthetic (Sec. 3)	[54]
Kimi k1.5 2025-01	Moonshot AI Multimodal reasoning	L2	L2 math/reasoning, knowledge, multimodal categories (App. B)	L2 FastText, OCR quality, LM filters, dedup (App. B)	Mix: L2 category-level corpus (App. B) Mid: L2 RL + long-context stages (Sec. 2)	Syn.: L2 synthetic multimodal/RL data (App. B/ Sec. 2)	[174]
MiniMax-01 2025-01	MiniMax General / long context / multimodal	L2	L2 academic literature/books/web/ code (Sec. 4.1.1)	L2 rule cleaning, dedup, reward-model quality scores (Sec. 4.1.1)	Mix: L2 balanced sampling by quality/ category (Sec. 4.1.1) Mid: L2 long-context pre-training (Sec. 4.3)	–	[175]
Phi-4-Mini 2025-02	Microsoft Small LM / multilingual	L2	L2 multilingual/code/math data (Sec. 3)	L2 filtering (Sec. 3)	Mix: L2 qualitative mixture (Sec. 3) Mid: L2 long-context/multimodal variants (Sec. 3)	Syn.: L2 synthetic data (Sec. 3)	[176]
Gemma 3 2025-03	Google Multimodal / multilingual	L2	L2 text/image/multilingual categories (Sec. 3)	L2 quality + safety filters (Sec. 3/Sec. 8)	Mix: L2 modality/language mixture (Sec. 3) Mid: L2 multimodal continued training (Sec. 3)	Syn.: L1 post-training synthetic data (Sec. 4)	[128]
LLaMA 4 Scout / Maverick 2025-04	Meta Multimodal / MoE	L2	L2 public + licensed + Meta-product data (model card)	L1 filtering/safety (model card)	Mid: L2 multimodal + long-context training (model card) Ann.: L1 cooldown/annealing (model card)	Syn.: L1 distillation/synthetic data (model card)	[145]
Qwen 3 2025-05	Alibaba Multilingual / reasoning	L3	L3 S1 >30T, S2 +5T, S3 long-context data (blog)	L3 VL-OCR, filtering, data selection (blog)	Mix: L3 S1/S2/S3 stages (blog) Mid: L3 reasoning + long-context stages (blog) Ann.: L2 annealing stage	Syn.: L3 math/code/reasoning synthetic data (blog)	[6]
MiniMax-M1 2025-06	MiniMax Reasoning / long context / RL	L2	L1 based on MiniMax-Text-01, no new base corpus (Abstract)	–	Mix: L2 diverse RL problems (Abstract) Mid: L2 CISPO RL, 40K/80K thinking budgets (Abstract)	Syn.: L2 RL on sandbox + real-world SWE environments (Abstract)	[177]
GLM-4.5 2025-07	Zhipu AI (Z.AI) Multilingual / reasoning / MoE	L3	L3 23T tokens (paper/abstract)	L2 filtering/scoring pipeline (Sec. data)	Mix: L2 pretrain + reasoning/code stages Mid: L2 agent/reasoning mid-training	Syn.: L2 reasoning synthetic data (Sec. data)	[146]
Kimi K2 2025-07	Moonshot AI Agentic / MoE	L3	L3 15.5T tokens, Web Text/Code/ Math/Knowledge (Sec. 2.2)	L2 K1.5 pipelines, correctness/ quality validation (Sec. 2.2)	Mix: L2 four pretrain domains (Sec. 2.2) Mid: L2 multi-stage post-training (Sec. 3) Ann.: L2 MuonClip, zero loss spike, LR schedule (Sec. 2)	Syn.: L3 post-train agentic data synthesis (Sec. 3)	[57]
DeepSeek V3.1 2025-08	DeepSeek General / long context / agent	L3	L3 840B-token continued pre-training (release note)	–	Mid: L2 long-context continuation (release note)	–	[178]
MiniMax-M2 2025-10	MiniMax Coding / agentic MoE	–	–	–	–	–	[179]

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Table VI. Evidence table for public disclosure of pretraining data in LLM technical reports and model cards (continued).

Model	Org. / Scope	Peak	Corpus Sources	Processing / Dedup.	Mixture and Training Schedule	Synthetic / Eval Evidence	Ref.
DeepSeek V3.2 2025-12	DeepSeek Reasoning / agent / sparse attention	L3	L1 V3.2-Exp/V3 lineage, no full base corpus (Sec. 1)	–	Mix: L2 reasoning + tool-use post-training (Sec. post-train) Mid: L2 scalable RL + agent synthesis (Sec. post-train)	Syn.: L3 1,800+ envs, 85k+ agent instructions (Sec. post-train)	[180]
GLM-5 2026-02	Zhipu AI (Z.AI) Multilingual / MoE / agent	L3	L3 28.5T tokens (Sec. 1/abstract)	L2 data filtering/scoring (Sec. data)	Mix: L2 ∼27T pretrain + mid-training (report) Mid: L2 long-context + agentic mid-training	Syn.: L2 reasoning + agent data (Sec. data)	[7]
Kimi K2.5 2026-02	Moonshot AI Multimodal / agentic	L3	L3 ∼15T mixed visual + text tokens (Sec. 4.3/T3)	L1 inherits K2/K1.5 pipelines (Sec. 4.3)	Mix: L3 ViT + joint pretrain + mid-training stages (Sec. 4.3/T3) Mid: L2 high-quality data + long-context activation (Sec. 4.3)	–	[147]
Qwen 3.5 / Qwen3.5-Omni 2026-02	Alibaba Native multimodal / multilingual	L3	L3 Omni S2 ∼4T tokens, text 0.92T, audio 1.99T, image 0.95T, video 0.14T, video-audio 0.29T (Sec. 3)	L1 stricter filtering (blog)	Mix: L3 modality token breakdown (Sec. 3) Mid: L2 S1 encoder alignment, S2 general, S3 long-context to 262K (Sec. 3)	–	[181, 182]
DeepSeek V4 2026-04	DeepSeek Long-context / reasoning / MoE	L3	L3 V4-Pro/Flash >32T tokens, 1M context (HF card)	L2 web filtering, V3 preprocessing, sample-level attention masking (report/HF card)	Mix: L2 math/code/web/long documents (report) Mid: L2 domain experts via SFT+GRPO, on-policy distillation (HF card) Ann.: L2 LR + long-context schedule (report)	Syn.: L2 agentic data in mid-training (report)	[8]
Kimi K2.6 2026-04	Moonshot AI Multimodal / coding / agentic	–	–	–	–	–	[183]